

Golden, CO (willing to relocate)
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Pratiksha Satish

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EDUCATION

COLORADO SCHOOL OF MINES

Bachelor of Science in Design Engineering | Minor in Business & Entrepreneurship

Expected: May 2027

SKILLS & STRENGTHS

Technical: SolidWorks, Fusion 360, Autodesk Inventor (Certified), Python, Java, EES, Vensim, Rhino, ArcGIS

Modeling & Prototyping: Systems modeling, Dynamic simulation, Thermodynamic performance analysis

Communication & Leadership: Dialogue facilitation, Workshop facilitation, Event planning, Product marketing, Cross-functional collaboration, Project management

EXPERIENCE

PEER EDUCATION

Peer Educator - Mental Health

Golden, CO
Aug 2025 - Present

- Led peer workshops and outreach events reaching 60+ students per event to promote mental health awareness
- Provided one-on-one support and connected students with appropriate campus wellness resources

PROJECTS

Indoor Air Quality Systems Model

Aug 2025 - Dec 2025

- Modeled indoor PM2.5 accumulation and identified filtration and ventilation as primary drivers of long-term exposure levels
- Showed MERV 16 filtration cut long-term performance decline by about 65% at no added annual cost
- Simulated emissions policy reducing outdoor PM2.5 from 9 to $7 \mu\text{p}/\text{m}^3$, reducing indoor particle load about 40%

Combined Cycle Power Plant Performance Model

Oct 2025 - Dec 2025

- Built an EES model of a combined-cycle plant to evaluate power and efficiency impacts of system upgrades
- Modeled turbine inlet temperature increase, improving output about 470 kW with about 2.0 ROI
- Increased plant efficiency through enhanced HRSG heat recovery without additional fuel input

PVC Flute

March 2025 - May 2025

- Designed and tuned a playable PVC flute in the key of A using recycled materials and iterative acoustic testing to demonstrate resourcefulness

LEADERSHIP & INVOLVEMENT

SOUTH ASIAN STUDENT ASSOCIATION (SASA)

Founder

Golden, CO
Sep 2024 - Present

- Founded and launched the first South Asian Student Association at Mines
- Promoted South Asian culture in the community by hosting multiple events successfully attracting 100+ attendees and establishing SASA as a recognized presence on campus
- Serving as President, leading initiatives, managing events, and overseeing expenses, while fostering collaboration with campus partners to promote diversity and inclusion

SOCIETY OF ASIAN SCIENTISTS & ENGINEERS (SASE)

Social Chair

Golden, CO
May 2025 - Present

- Organize and lead 1–2 events per month and bi-weekly general meetings to foster an inclusive community while promoting member networking and professional development
- Expanded outreach to engage a broader audience, creating opportunities for networking among Asian and non-Asian members through diverse, inclusive activities

RELEVANT COURSEWORK:

Thermodynamics & Energy Systems (cycle analysis) · Systems Modeling & Simulation (stock-flow models, feedback loops) · Human-Centered Design (anthropometrics, inclusive design) · Physical Prototyping & Fabrication(arduino systems, iterative testing) · GIS & Spatial Analysis(environmental mapping)

INTERESTS

Language: English, French, Tamil

Activities: Music, Dance, Graphic Design, Cooking, Composition